30066 – MACROECONOMICS

(8 CFU)

CLASS : BIEM 14 & 19
2024 – 25

<u>Lecture 15</u>
(Ch. 9)

The *IS-LM-PC* model: demand and supply shocks

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WHERE WE ARE

Fundamentals

1. Macroeconomic variables
   – Elements of national accounting

2. The short run
   1. Goods markets
   2. Financial markets
      1. Without commercial banks
      2. With commercial banks
   3. IS-LM model

3. The medium run
   1. Labor market
   2. Phillips curve
   3. IS-LM-PC model
      1. Medium and short run equilibrium
      2. Demand and supply shock

Extensions

4. Expectations, financial markets, and economic policies
   1. Financial markets and expectations
   2. Consumption, investment, and expectations
   3. Output, economic policies, and expectations

5. Open economy
   1. Exchange rates, trade, and international returns
   2. Goods markets in an open economy
   3. Financial markets and economic policies in an open economy

6. Public debt and fiscal policy

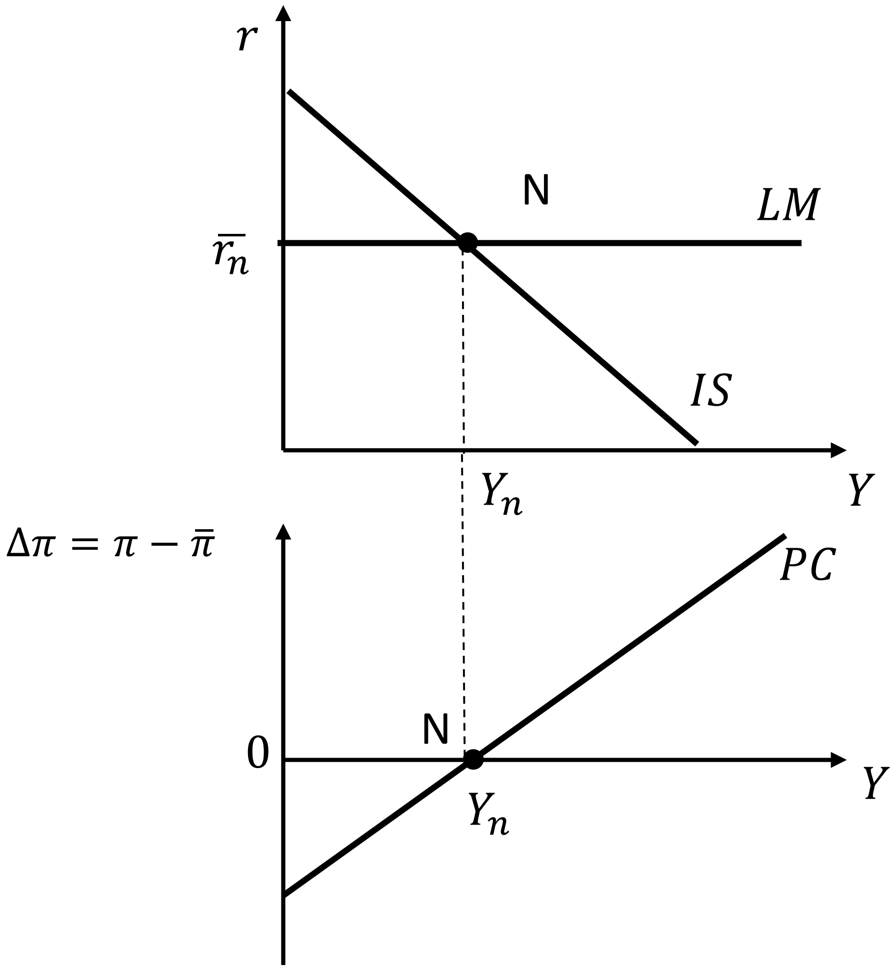
7. From the Great Recession to Quantitative Tightening
   – From a housing crisis to a financial crisis
   – Unconventional monetary policies

Expectations are well anchored to the CB target $\pi^e = \bar{\pi}$

$$IS: \quad Y = C(Y - T) + I(Y, r + x) + G$$

$$LM: \quad r = \bar{r}$$

$$PC: \quad \Delta\bar{\pi} = \frac{\alpha}{L}(Y - Y_n)$$



The economy is in a medium-run equilibrium: point N

$IS \cap LM$ at Y_n

$Y = Y_n, u = u_n$

Inflation is at the target

$\Delta\bar{\pi} = 0 \leftrightarrow \pi = \bar{\pi}$

MEDIUM RUN EQUILIBRIUM

All markets (goods, financial, labor) are in equilibrium and expectations are correct => the economy is at its potential with no output gap

$$Y = Y_n, u = u_n, \Delta\pi^e = 0$$

Expectations are well anchored to the CB target $\pi^e = \bar{\pi}$

Shocks can hit the economy

SHORT RUN SHOCKS

Demand shocks: no effect on Y_n and real effects are only temporary

Supply shocks: effect on Y_n and produce permanent effects also in the medium run

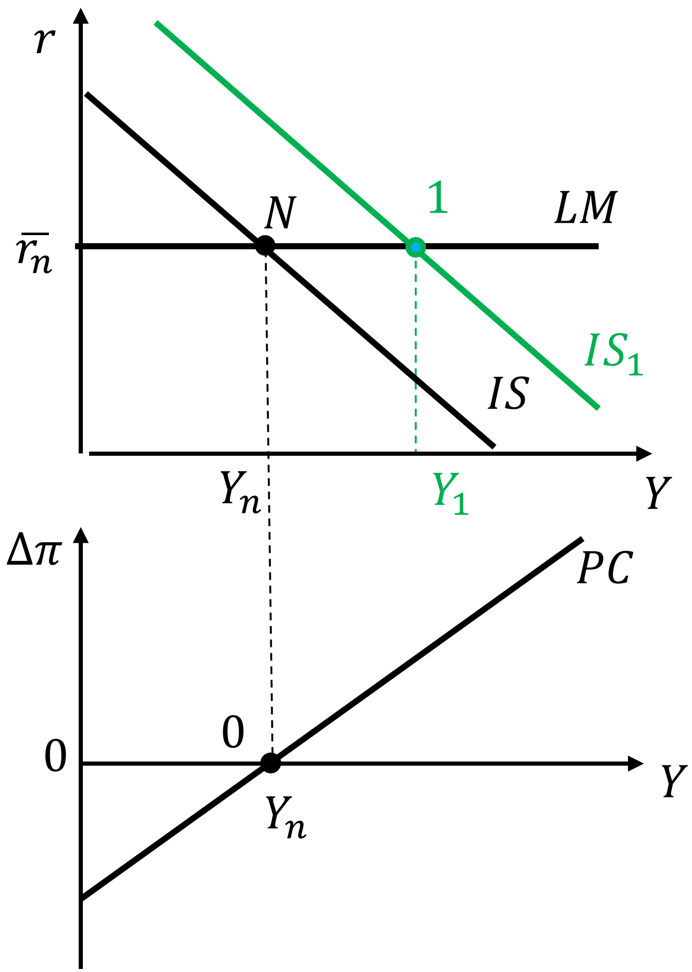
FISCAL POLICY

The Government implements an expansionary fiscal policy investing in public schools

The CB keeps the interest rate constant

- What is the impact of such an intervention in the short run and in the medium run?
- What about the level and the composition of demand?

Initial medium-run equilibrium: point N



Government invests in public schools
Fiscal expansion: $G \uparrow$

- $G \uparrow \Rightarrow Z \uparrow \Rightarrow Y \uparrow \Rightarrow \quad C\&I \uparrow \Rightarrow Y \uparrow$: IS right
- LM does not change

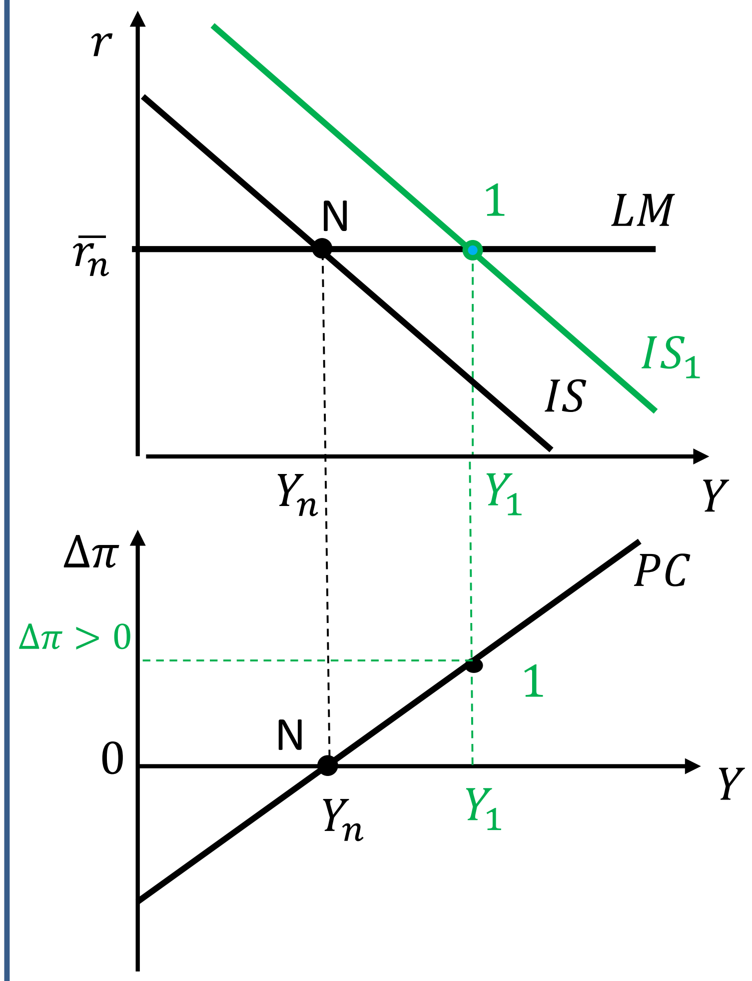
Short-run equilibrium: 1
The economy goes above its potential
$Y \uparrow, i =, \bar{r} =$

What happens to the inflation rate?

In the short run the economy moves from N → 1 (short run equilibrium)

Government invests in public schools

Fiscal expansion: $G \uparrow$



What happens to the inflation rate?

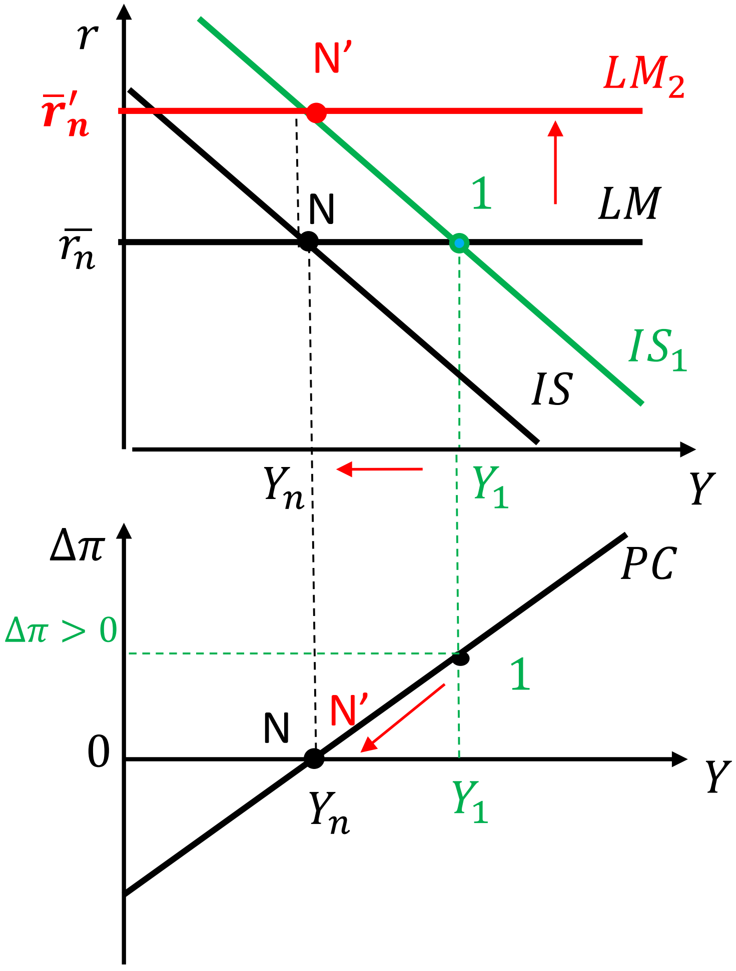
Since $u \downarrow$, W growth $\uparrow \Rightarrow \quad \Delta\pi > 0$

Current inflation is higher than expected

Inflation is above the CB target but constant:

$\pi > \bar{\pi}$

How can the CB bring inflation back to the target?



To control inflation, the CB must increase nominal interest rate so that also the real one increases

LM up

i ↑ and r ↑ ⇒ I ↓ ⇒ Z ↓ Y ↓: along the IS

Medium-run equilibrium: N'

$Y = Y_n, \qquad r = \bar{r}_n', \qquad \Delta\pi = 0$

Inflation at the CB target : $\pi = \bar{\pi}$

EFFECTS

SHORT RUN

- $Y \uparrow, \bar{\imath} =, \bar{r} =, u \downarrow$
- $\Delta\pi > 0, \pi > \bar{\pi}$
- π constant above the CB target

MEDIUM RUN

- $Y = Y_n, u = u_n, \bar{\boldsymbol{r}}'_{\mathbf{n}}$
- $\Delta\pi = 0, \pi = \bar{\pi}$
- π constant at the CB target

Is fiscal policy neutral in the medium run?
Do components of aggregate demand change?

ECONOMIC INTUITION

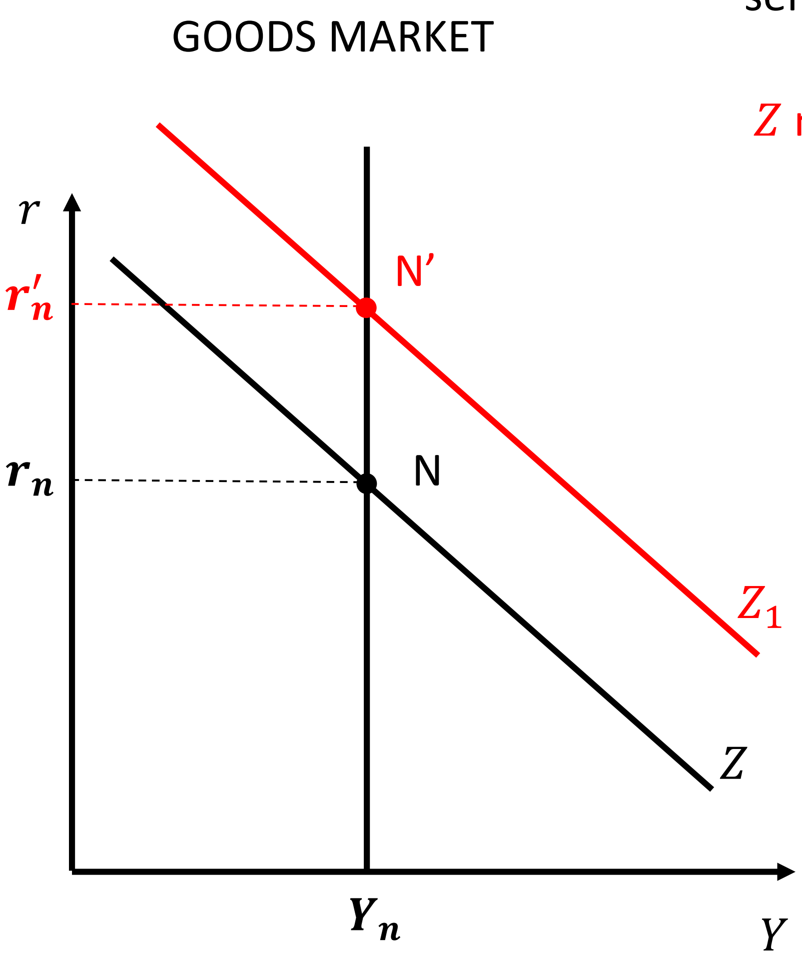
SHORT RUN

- In the short run, on the goods market, the increase in public expenditure increases demand and, through the multiplier, income.
- The higher income in the financial markets increases money demand given the higher level of real transactions for goods an services. To keep the interest constant the CB increases the money supply.
  - The nominal interest rate dos not change and, given expected inflation, also the real one stays constant.
- The increase in output, increases employment and reduces unemployment.
  - Workers have more bargaining power and nominal wages grow as well as firms production costs.
  - To maximize their profits firms charge higher prices and inflation raises above the CB target.

MEDIUM RUN

- To keep inflation at the target level over the medium run the CB has to intervene and, in order to close the positive output gap, a contractionary monetary policy has to be implemented
  - The CB sells bonds, on the bonds market the price of bond decreases and, given their face value, their nominal return increases.
  - The higher policy rate, given expectations, increases the real interest rate
- The higher borrowing rate reduces investment, the aggregate demand and through the multiplier income
- The decrease in GDP has an effect on the labor market where workers bargaining power with respect to firms decreases, since employment decreases and unemployment increases
  - Bargained nominal wages decrease thus reducing firms production costs.
  - Firms can maximize their profits by charging lower prices
  - The inflation rate decreases up to the CB target
- The economy is back at its potential with no output gap

IMPACT OF FISCAL POLICY IN THE MEDIUM RUN



$G\uparrow$: increase in demand of goods and services

Z moves to the right

$Y = Y_n \quad BUT \quad r_n \uparrow$

IMPACT OF FISCAL POLICY IN THE MEDIUM RUN

In the new medium-run equilibrium:

$$Y = Y_n \quad and \quad r_n \uparrow$$

$$Y_n = \boxed{C(Y_n - T) + I(Y_n, r_n + x) + G}$$

Z_n = aggregate demand

Since Y_n constant $\Rightarrow$ constant level of aggregate demand

BUT

The **composition** of demand is **different** since:

$$C =$$
$$I \downarrow$$
$$G \uparrow$$

*Fiscal policy is **not neutral** in the medium run:*
it has real effects even in the medium run

SUMMARY

Impact of an expansionary fiscal policy

- Only temporary effect on income
- But permanent effect on the composition of demand (non neutrality in the medium run)

SUPPLY SHOCKS

Structural shocks that change permanently the potential output
=> Structural reforms of the economy that change

- Competition
- Labor market institutions
- Productivity
- Costs of production

…

Affect the PC

OIL SHOCK

Sudden surge in oil prices

Assumption: no direct impact on demand for goods and services
The CB keeps the interest rate constant

- What is the impact of such an intervention in the short run and in the medium run?
- What about the level and the composition of demand?

THE OIL SHOCK

- Oil is a raw material for all the firms
- In our model we do not consider the cost of raw materials
- How can we represent this shock in our model?
- If the price of raw materials increases:
  - The marginal production cost increases, given wages
  - Firms increase prices
  - We can say that the parameter $\boldsymbol{m}$ **increases**

The equilibrium of the labor market changes: u_n and Y_n change

Structural change

IMPACT ON THE LABOR MARKET

$$WS: \frac{W}{P} = F(\underset{-}{u}, \underset{+}{z})$$

$$PS: \frac{W}{P} = \frac{A}{1+m}$$

$m \uparrow$

Which curve is affected?

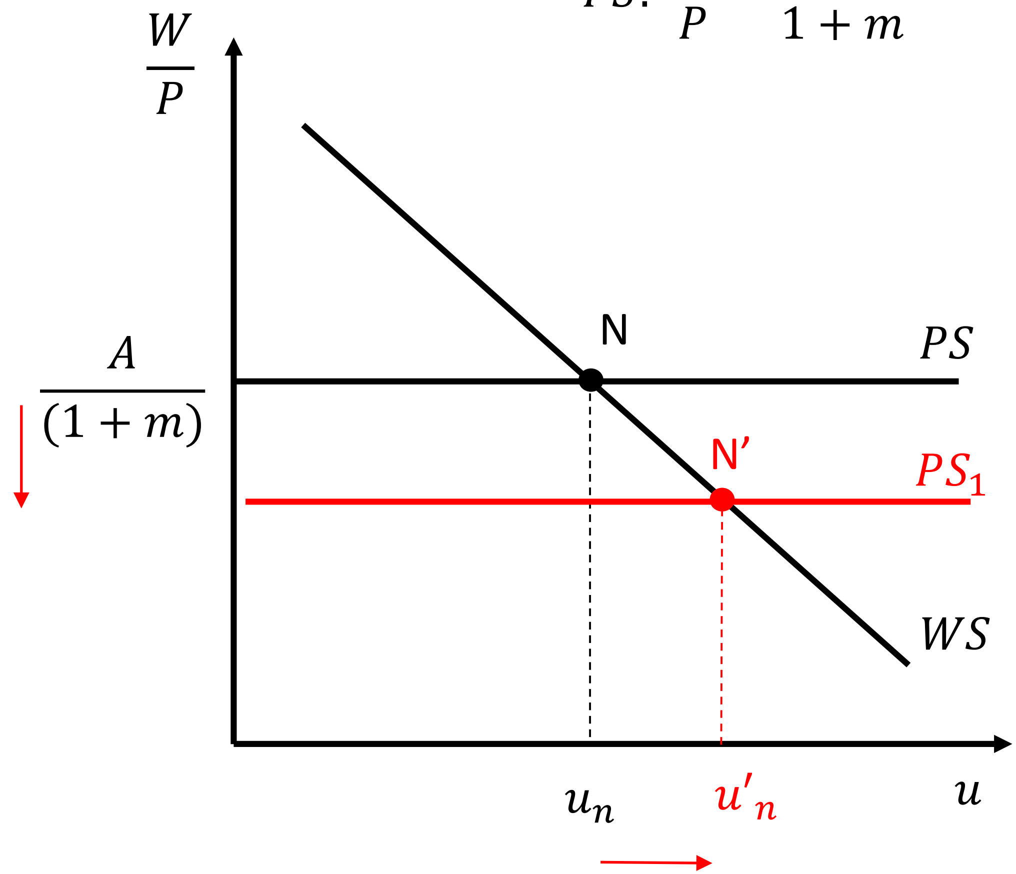
PS down

$\text{m} \uparrow \Rightarrow$ given W, $\text{MC} \uparrow$

$\Rightarrow \text{P} \uparrow \Rightarrow \downarrow \frac{\text{W}}{\text{P}}$ paid firms to max profits

More frictions in the labor market and to induce workers to accept a lower real wage $\boldsymbol{u_n} \uparrow$

and $\boldsymbol{Y_n} \downarrow$



NEW POTENTIAL LEVELS

- The natural rate of unemployment rises: $u_n \uparrow$

$$u_n' > u_n \uparrow$$

- Potential GDP declines: $Y_n \downarrow$

$$Y_n' < Y_n$$

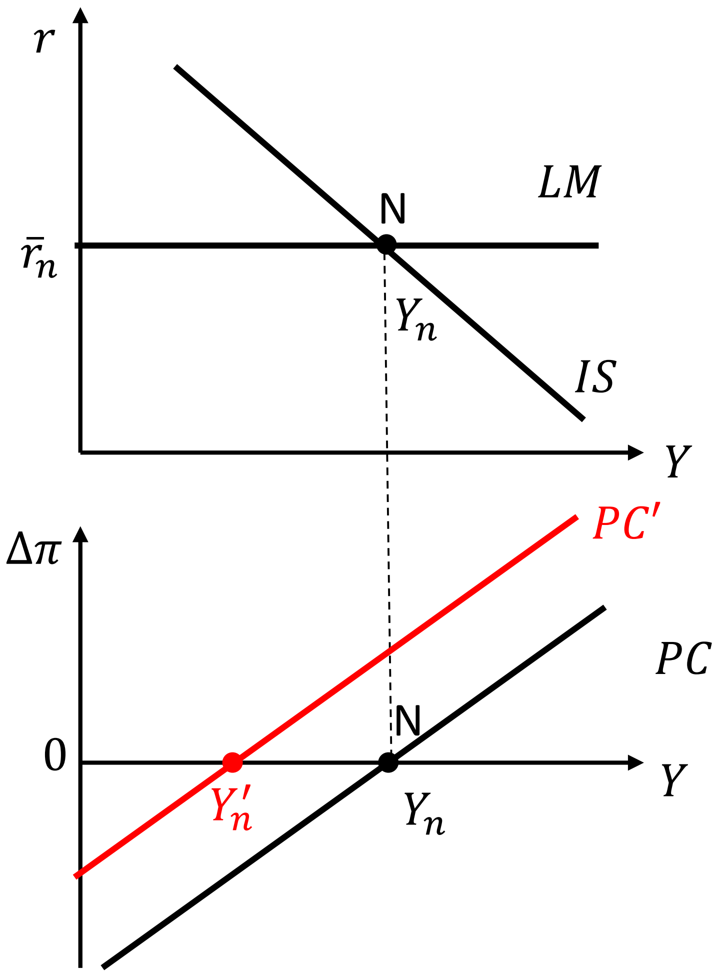
- No effect on actual output

…BUT…
a positive output gap arises:
the economy goes above the new potential

Which curve is affected in the IS-LM – PC graph?

Initial medium-run equilibrium: point 0

$m \uparrow$ $\quad$ $\text{P} \uparrow (\pi \uparrow) Y_n \downarrow$



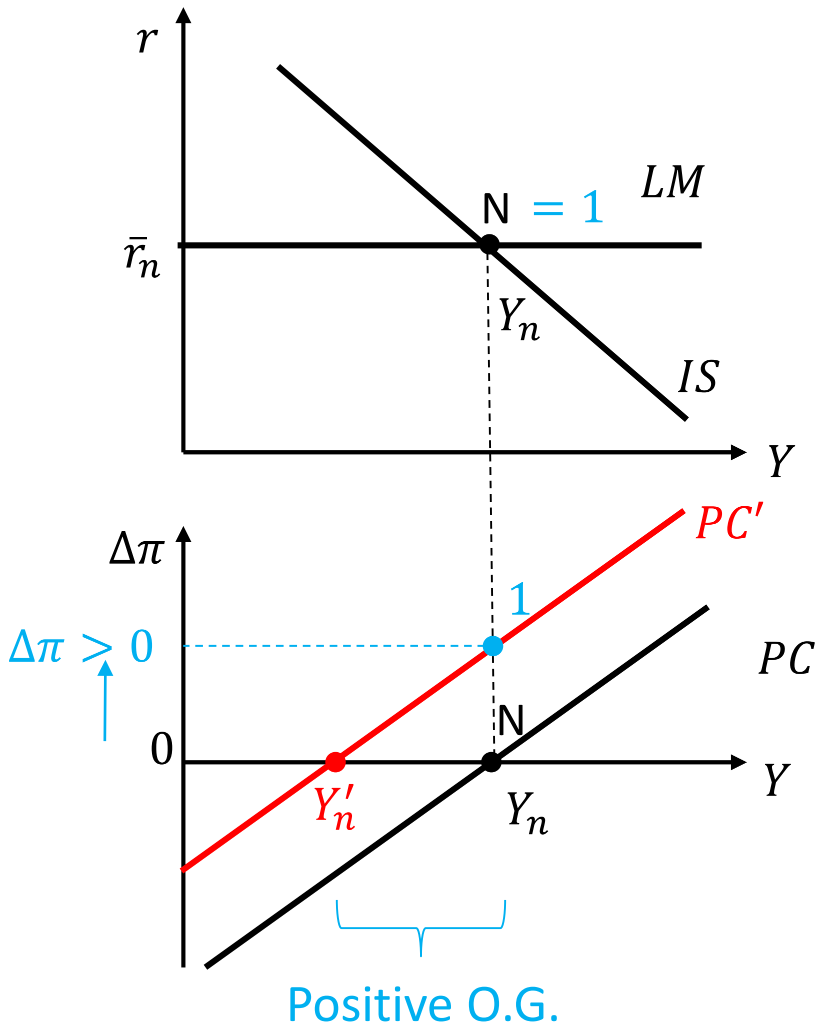
$$IS: \quad Y = C(Y-T) + I(Y, r+x) + G$$

$$LM: \quad r = \bar{r}$$

$$PC: \quad \Delta\pi = \frac{\alpha}{L}\left(Y - \boldsymbol{Y'_n}\right)$$

the PC shifts left/up

NEGATIVE SUPPLY SHOCK: short run



What happens to the short run equilibrium?

IS and LM don't move:
$point\ 1 \equiv point\ N$

Y does not change

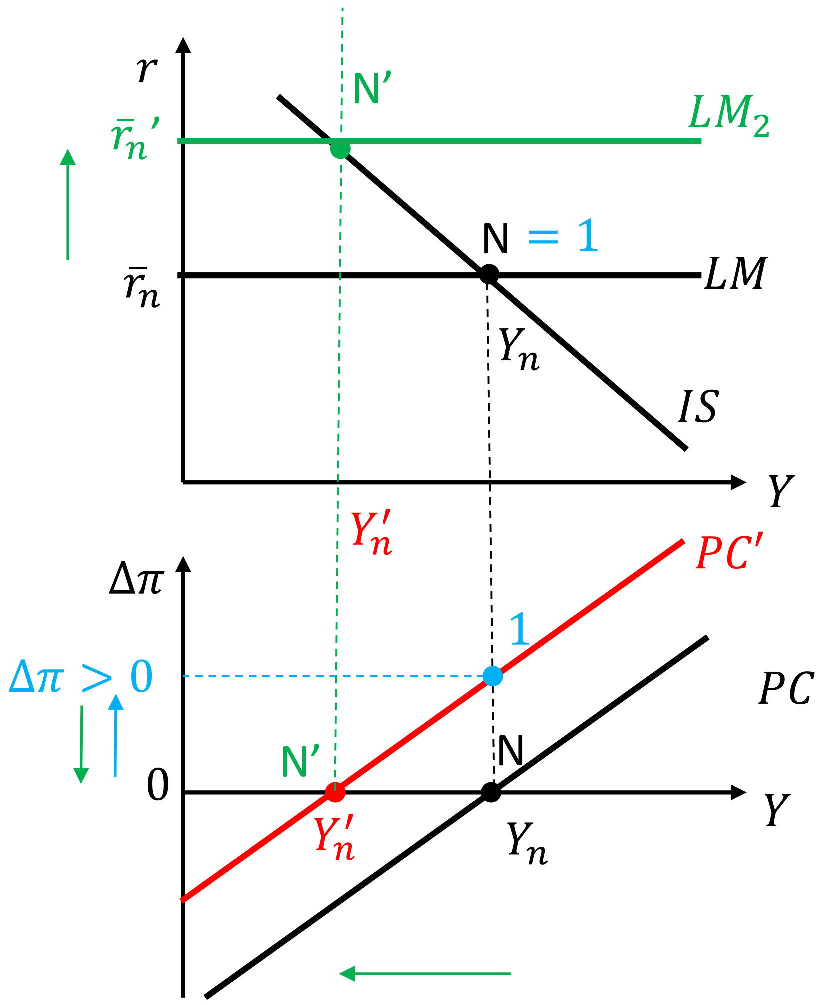
BUT

$Y_1 > Y_n' \Rightarrow \Delta\pi > 0$

Inflation increases and goes above the CB target: $\pi > \bar{\pi}$

1 = short run equilibrium

Adjustment to the MR



How does the economy adjust to the ***medium run****?*

The CB must increase *i* to increase r in order to reduce Y up to Y_n'

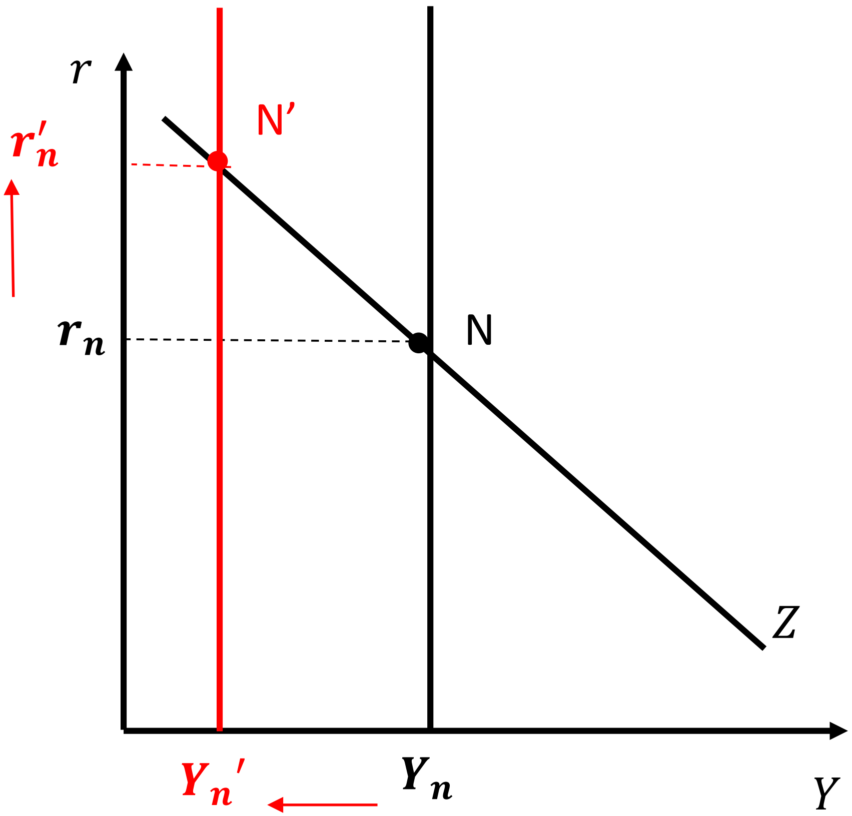
Medium-run equilibrium:N'

$\downarrow Y = Y_n{}' \quad \uparrow r = \bar{r}_n{}' \quad \Delta\pi = 0$

Inflation at the CB target : $\pi = \bar{\pi}$

What happens to r_n?
We need to study the Zn Yn graph

GOODS MARKET



The structural shock decreases Y_n

As a result $r_n \uparrow$ since on the goods market there is an excess demand and to clear the market is necessary to reduce demand in the medium run

IMPACT OF FISCAL POLICY IN THE MEDIUM RUN

In the new medium-run equilibrium:

$$Y_n \downarrow \quad and \quad r_n \uparrow$$

$$Y_n = \boxed{C(Y_n - T) + I(Y_n, r_n + x) + G}$$

Z_n = aggregate demand

Since Y_n decrease $\Rightarrow$ also the level of aggregate demand decreases

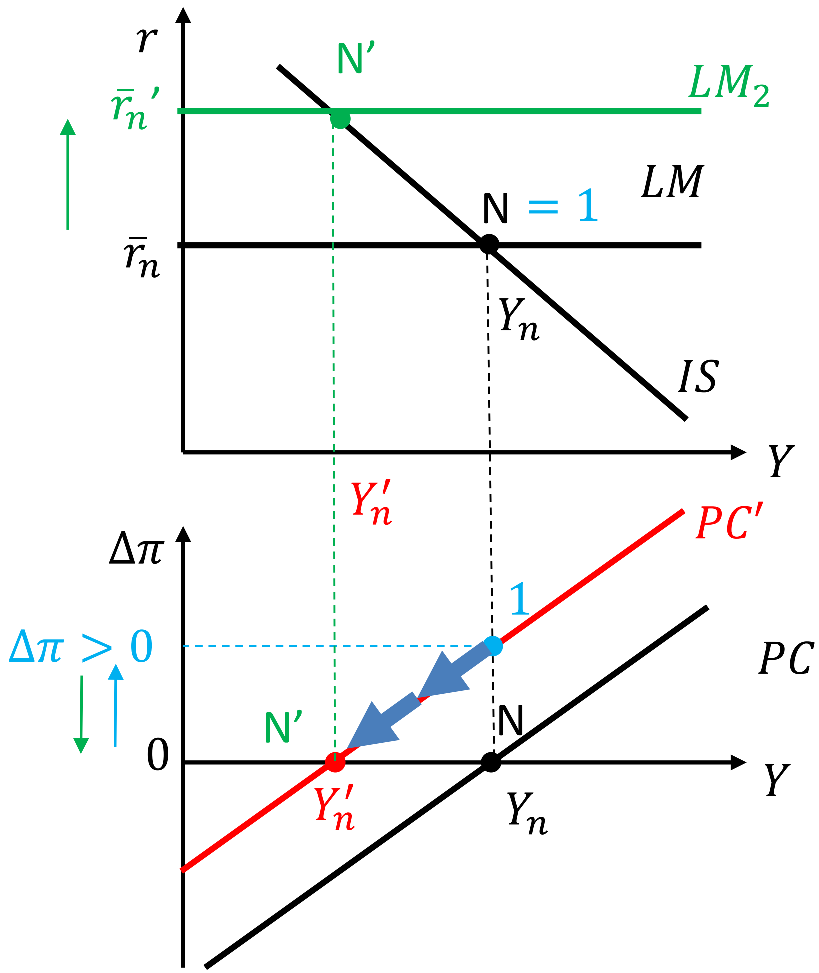
AND

The **composition** of demand is **different** since:

$$C \downarrow$$
$$I \downarrow$$
$$G =$$

*Supply shock is **not neutral** in the medium run: it has real permanent effects on level and composition*

NEGATIVE SUPPLY SHOCK: adjustment to the MR



In the reality the adjustment is gradual

What happens during the adjustment process?

$Y \downarrow$ & $\Delta\pi > 0$

STAGFLATION!!!

ECONOMIC INTUITION

SHORT RUN

- In the short run, on the products market, firms charge higher prices. This reduces the real wage that maximizes their profits.
- There is a positive gap between the real wage asked by workers and the one firms are willing to pay and this increases frictions in the labor market.
  - Unemployed workers look for a job less intensively
  - The unemployment rate in the medium run increases while potential output decreases

- Current output does not change, but the economy is above the new potential.
- The price level is higher in the short run as well as the inflation rate that goes above the CB target.

MEDIUM RUN

- To keep inflation at the target level over the medium run the CB has to intervene and, in order to close the positive output gap, a contractionary monetary policy has to be implemented
  - The CB sells bonds, on the bonds market the price of bond decreases and, given their face value, their nominal return increases.
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- The higher borrowing rate reduces investment, the aggregate demand and through the multiplier income

- The decrease in GDP has an effect on the labor market where workers bargaining power with respect to firms decreases, since employment decreases and unemployment increases
  - Bargained nominal wages decrease thus reducing firms production costs.
  - Firms can maximize their profits by charging lower prices
  - The inflation rate decreases up to the CB target

- The economy is back at its new lower potential with no output gap

SUMMARY

Impact of a negative supply shock

- Actual income does not change but current inflation changes

- Structural changes in the labor market that are permanent

- Potential income changes but in the medium rum inflation is bring back to the target and expectation are correct

- Permanent effect on the composition of demand (non neutrality in the medium run)

DEMAND VS. SUPPLY SHOCKS

- Demand shocks: only temporary effects in the short run since in the medium run income does not change
  - Positive demand shock in SR , in MR
  - Negative demand shock in SR , in MR

- Supply shocks: also permanent effects since income changes in the medium run
  - Negative supply shock in SR , in MR
  - Positive supply shock in SR $Y=$, in MR

IS-LM-PC: *ZLB AND DEFLATION SPIRAL*

Why do Central Banks target a positive inflation rate?

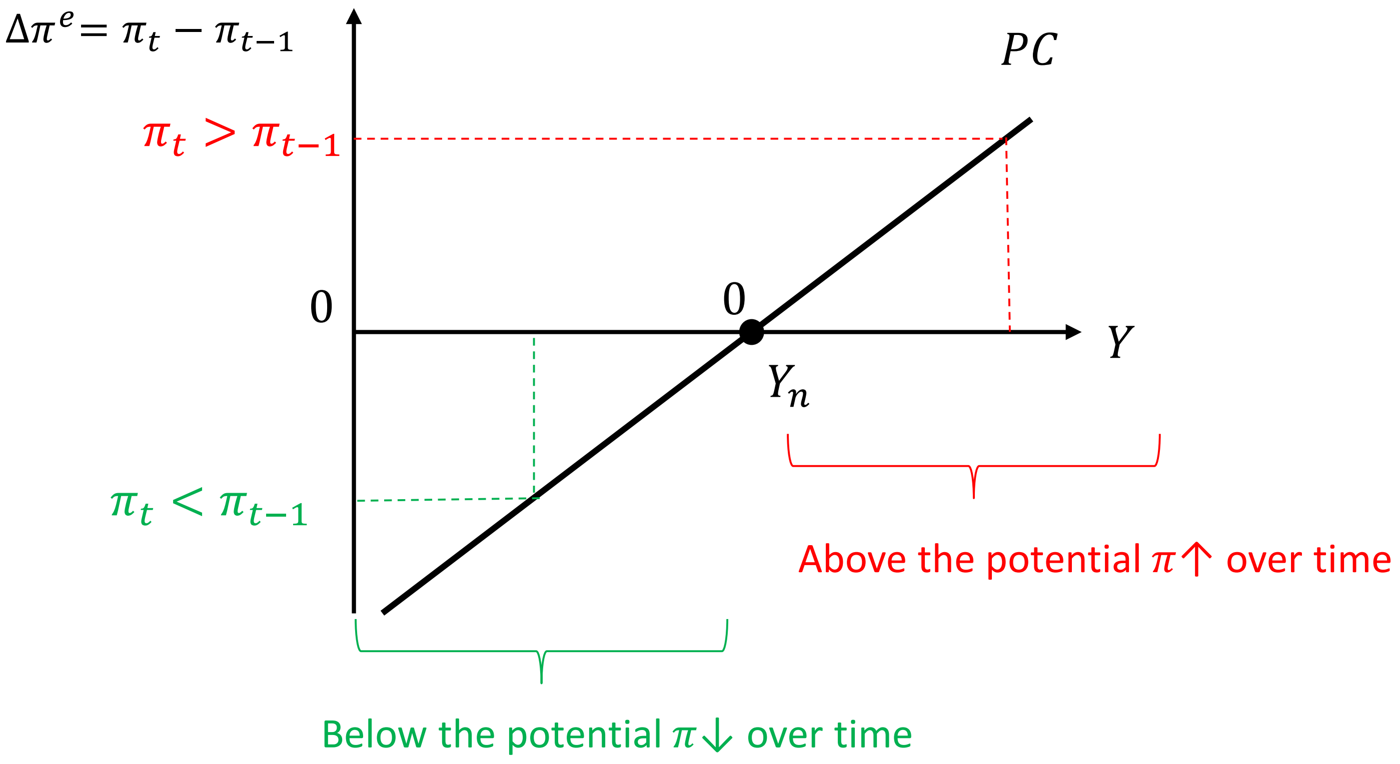
1. Constraints on monetary policy when the economy is at the *zero lower bound*

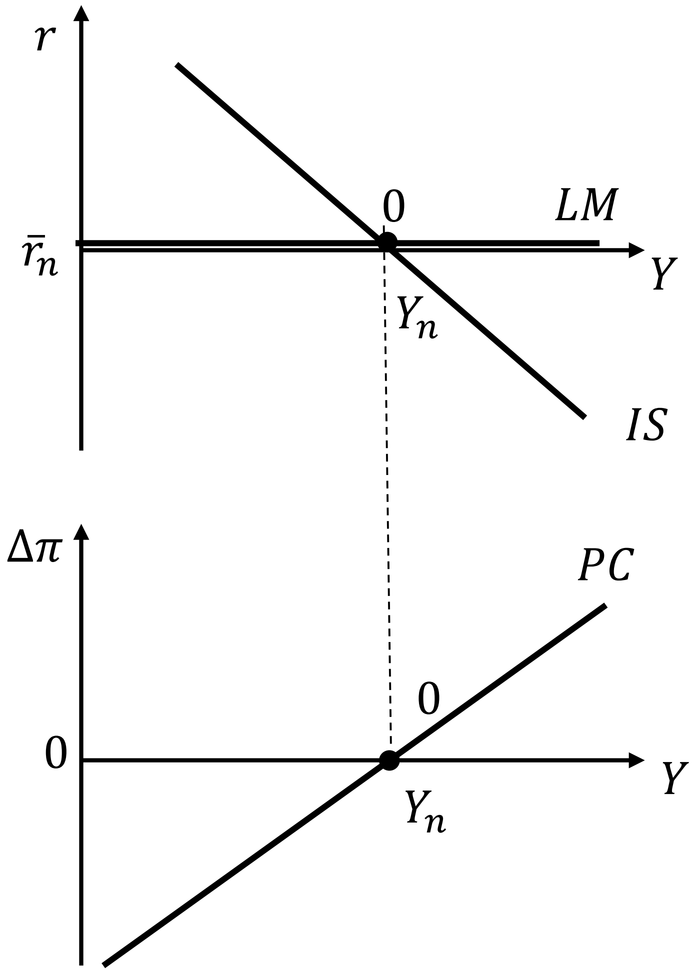
2. To avoid deflation spiral or deflation trap

- Imagine a very extreme situation
  - The economy is in a medium-run equilibrium at the *zero-lower bound* with no inflation
  - **Expectations are fully adaptive**

$$i = 0 \quad \pi = 0 \quad \pi^{e} = \pi_{t-1}$$

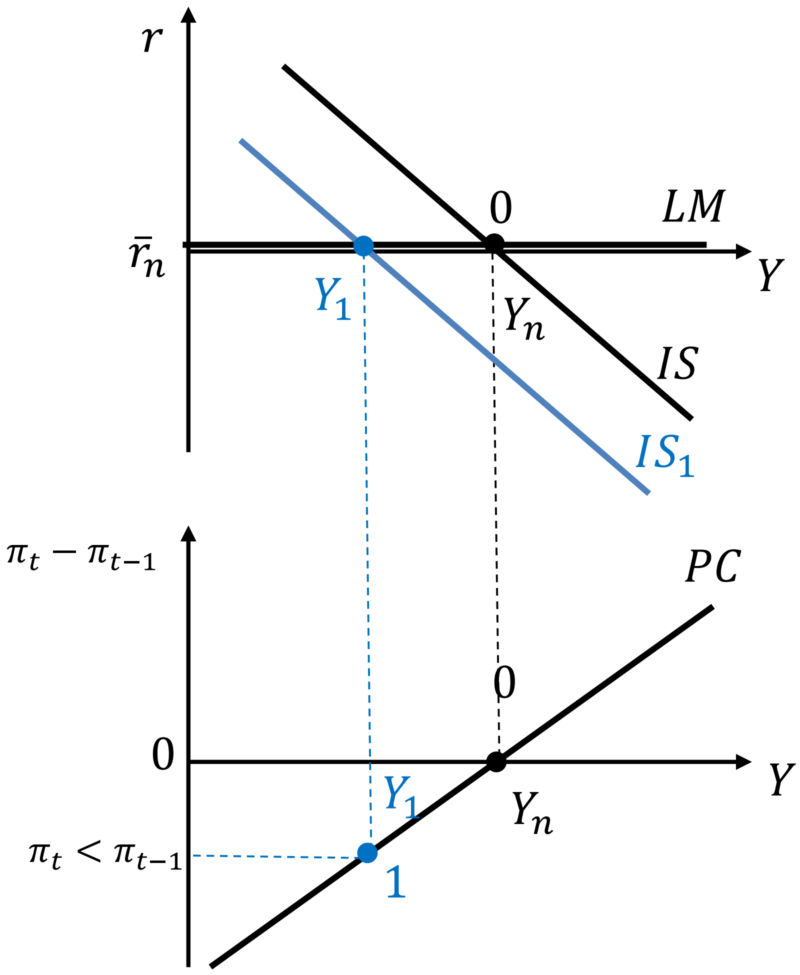
- What happens if a negative demand shock hits the economy?





r
0
LM
$\bar{r}_n$
Y
Y_n
IS
Δπ
PC
0
0
Y
Y_n

Initial medium-run equilibrium: point 0



Banking system crisis: $x \uparrow$
IS to LEFT: $Y \downarrow$ recession

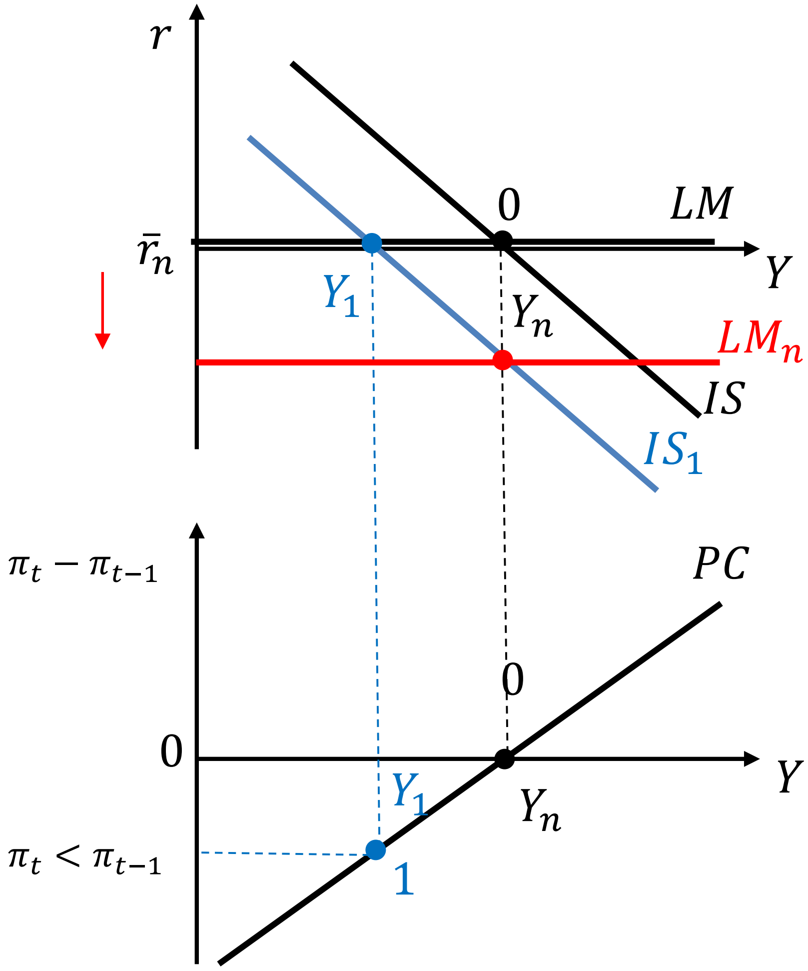
$$Y_1 < Y_n \Rightarrow u_1 > u_n$$

$\Rightarrow$ W growth $\downarrow \Rightarrow \Delta\pi < 0$

Given $\pi_0 = 0$, $\pi < 0$

DEFLATION!!!

What can the CB do to fight against the recession?
Can the CB restore the medium run equilibrium?



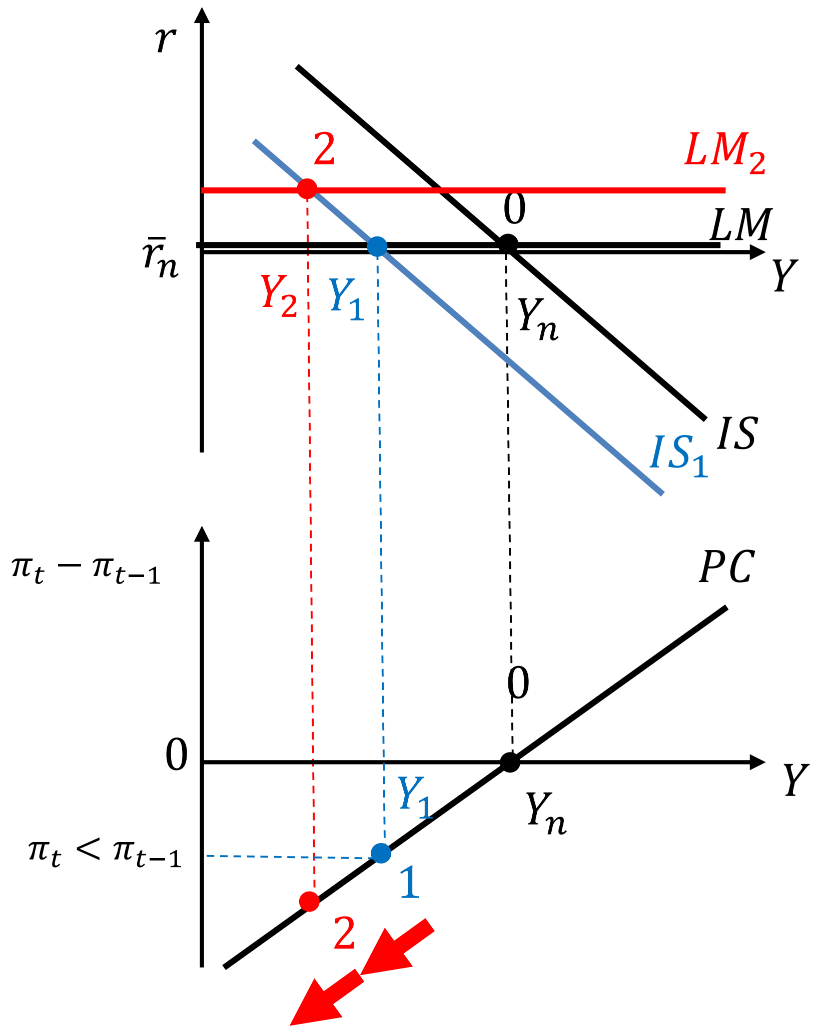
In principle, the CB should move the LM down to intersect the IS where $Y = Y_n$

Reduce i to reduce r

BUT the CB can't: $i = 0$

Conventional MP cannot generate an increase in GDP towards Y_n

Role of expectations: $r \downarrow$ if agents expected inflation



Conventional MP fails to move the economy to Y_n

If the CB cannot intervene, what happens with adaptive expectations and $\pi_0 = 0$?

Since $\pi < 0: \pi^e \downarrow \Rightarrow \pi^e < 0$

$$r = i - \pi^e = 0 - \pi^e > 0$$

The LM shifts **upward** also in the CB does not do anything

DEFLATION SPIRAL or TRAP

ZLB: Summary

- If the economy is at the *zero lower bound,* conventional MP cannot have expansionary impact

- In case of **deflation**, the economy may enter a **deflation spiral or trap**
  - The recession becomes more severe!!!

- What can we do?
  - Fiscal policy (but this has constraints too…)
  - Unconventional monetary policies
  - Well anchored expectations

MONETARY POLICY AND NEUTRALITY OF MONEY

The CB cannot push the economy above/below the potential forever

Assume

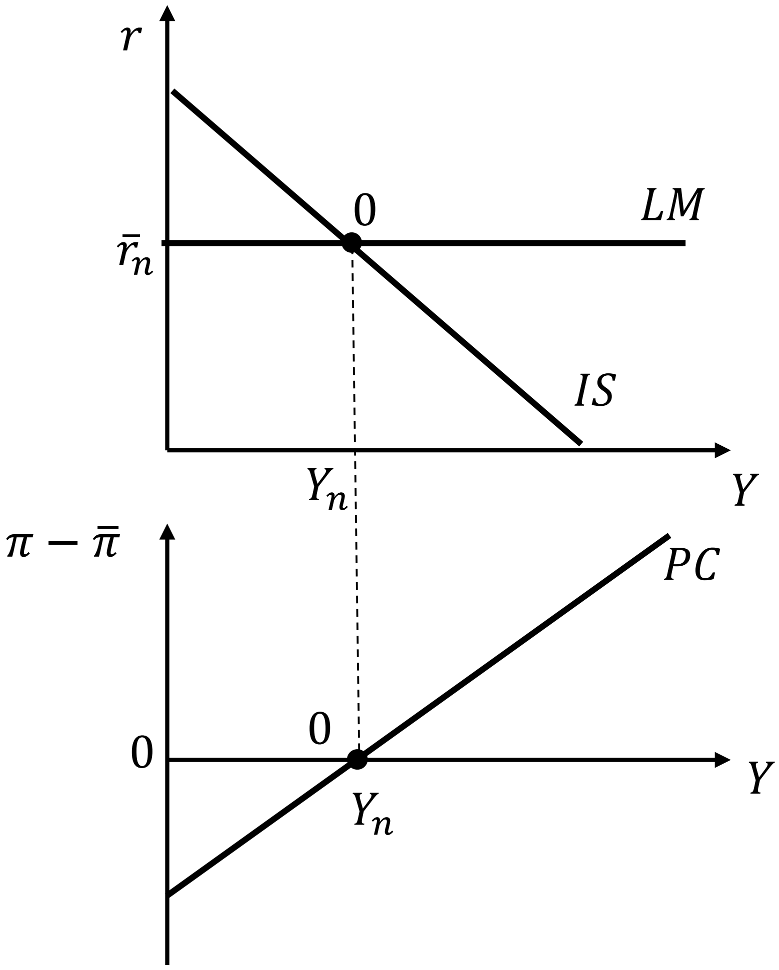
- An economy at the medium run equilibrium

- The Central Bank implements an expansionary monetary policy

- Anchored expectations
  - (not relevant for main results)

What is the impact in the short run? And in the medium run?

IS-LM-PC: STARTING EQUILIBRIUM



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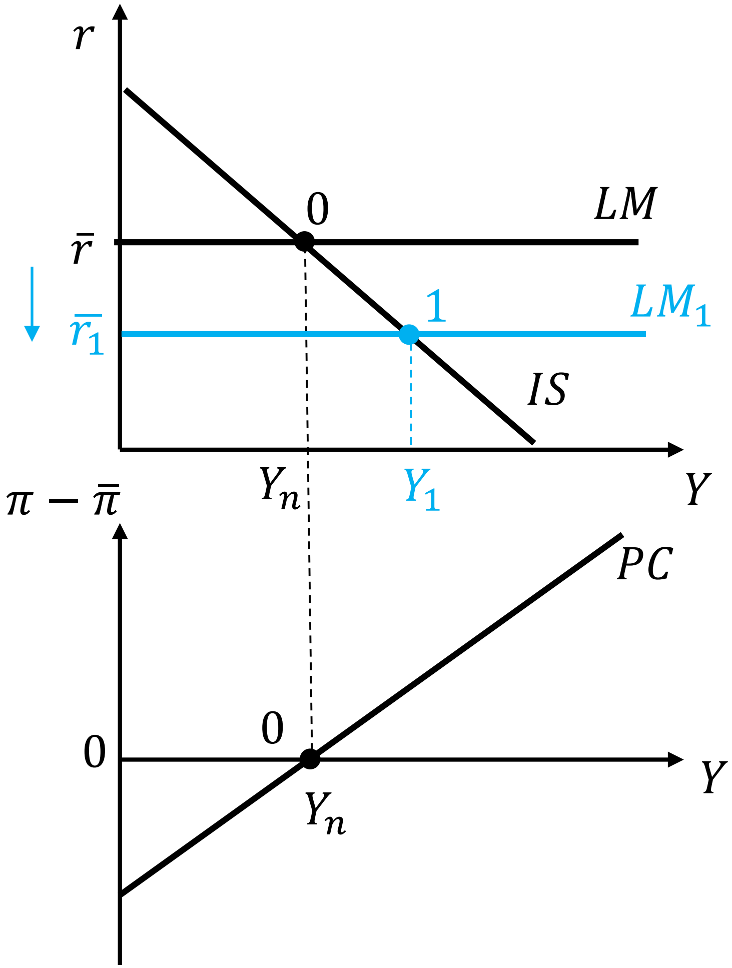
$$LM: \quad r = \bar{r}$$

$$PC: \quad \pi - \bar{\pi} = \frac{\alpha}{L}(Y - Y_n)$$

The economy is in a medium run equilibrium:
$Y = Y_n; \;\; \pi = \bar{\pi}$

IS-LM-PC: SHORT RUN



Expansionary monetary policy:

CB buys bonds: $M \uparrow \Rightarrow \bar{\imath} \downarrow$ and $\bar{r} \downarrow$

$\boldsymbol{LM} \downarrow$

$\bar{r} \downarrow \Rightarrow I \uparrow \Rightarrow Y \uparrow \Rightarrow C\&I \uparrow \Rightarrow Y \uparrow$

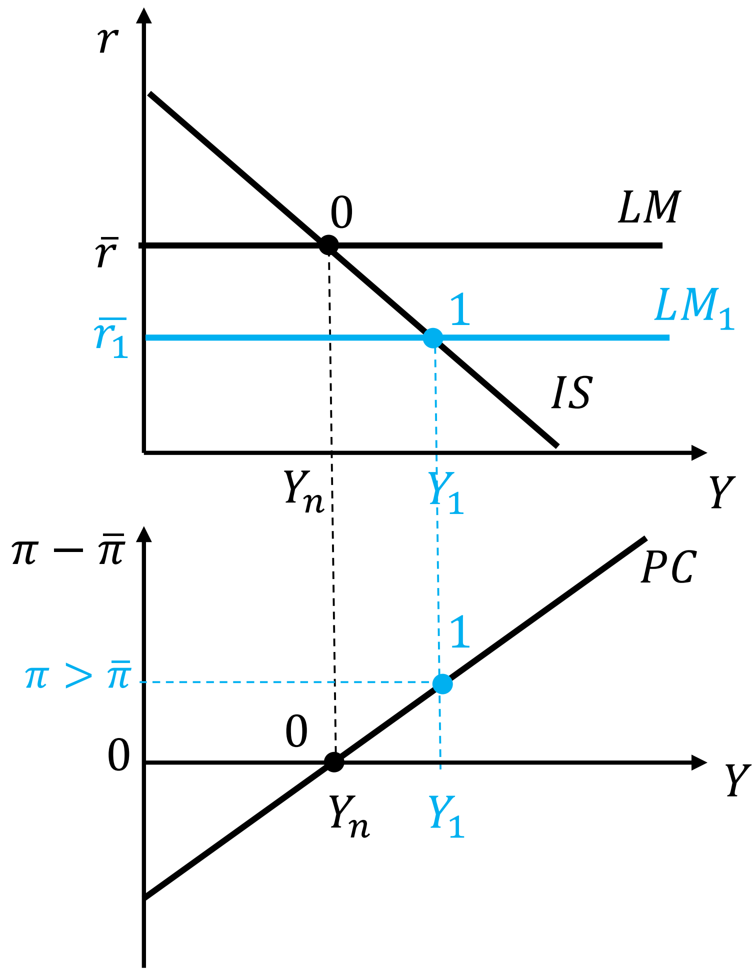
Along the IS

Short-run equilibrium: 1

$Y \uparrow; \quad \bar{r} \downarrow$

What happens to the inflation rate?

IS-LM-PC: SHORT RUN



Short-run equilibrium: 1

$Y \uparrow$; $\bar{r} \downarrow$

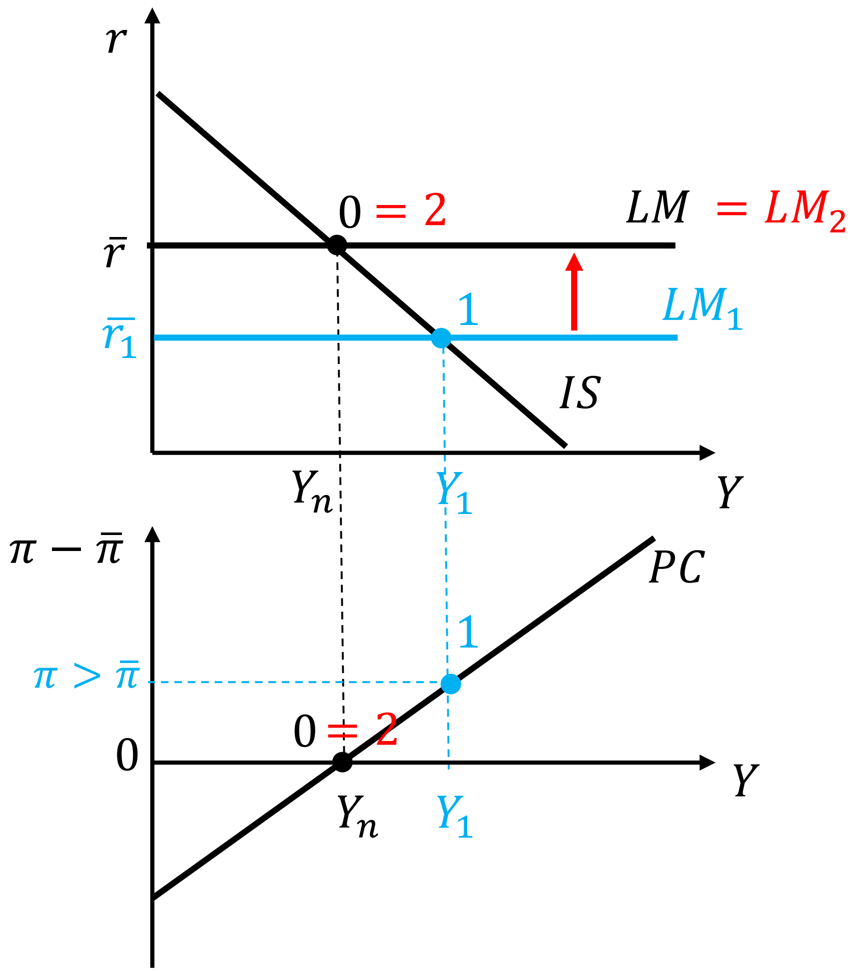
$Y > Y_n \Rightarrow u < u_n$:

W growth $\uparrow \Rightarrow \pi > \bar{\pi}$

Inflation goes above the CB target

How can the economy restore a medium run equilibrium?

IS-LM-PC: MEDIUM RUN



Adjustment to the medium run?

To close the output gap and bring inflation back to the target the CB must increase r through i

$$i \uparrow r \uparrow \Rightarrow I \downarrow \Rightarrow Y \downarrow$$

<u>LM goes back to the original position</u>

Medium-run equilibrium:

$$Y = Y_n, \qquad r = r_n, \qquad \pi = \bar{\pi}$$

IS-LM-PC: EXPANSIONARY MP

- **Short run**

$$Y \uparrow \quad \bar{r} \downarrow \quad \pi > \bar{\pi}$$

- **Medium run**

$$Y = Y_n, \; r = r_n, \; \pi = \bar{\pi} \text{ (initial value!)}$$

Note: if expectations were **adaptive inflation will be** constant, but at a higher value !!!

  – Between 0 and 2, no change in AD components

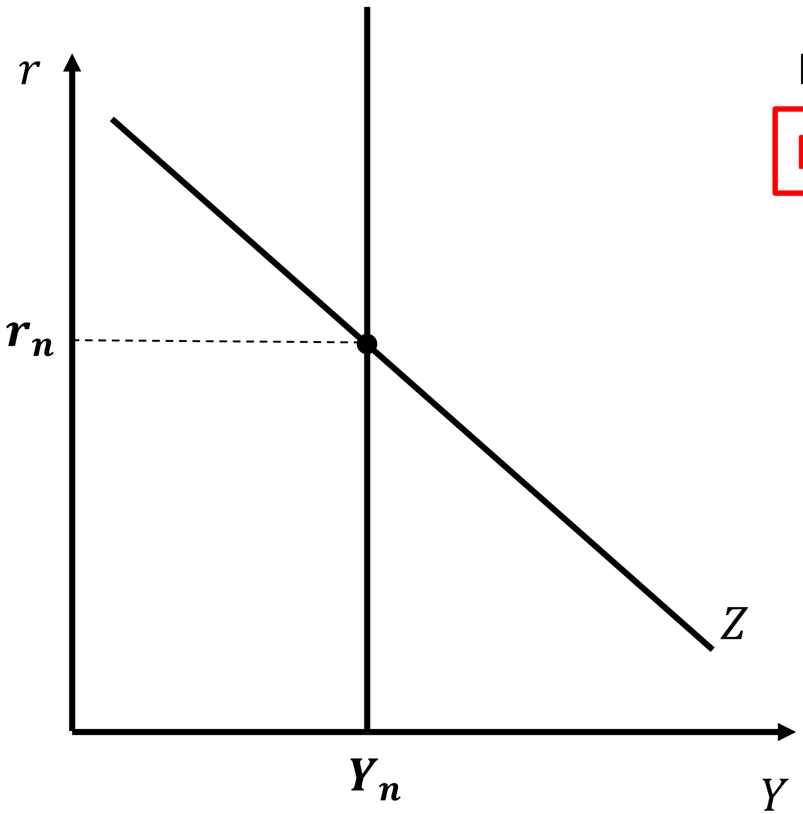
*No impact of monetary policy on **real** variables in the medium run*

Neutrality of money

Medium-run equilibrium:

$$\boldsymbol{Y_n} = C(\boldsymbol{Y_n} - T) + I(\boldsymbol{Y_n}, \boldsymbol{r_n} + x) + G$$

GOODS MARKET



No shifts in demand and supply

Nor Y_n and $\bar{r}_n$ change in the MR

MONEY is NEUTRAL

Monetary policy: **NO real effects in the MR**

- Only one variable affected: price level
- π has grown during the SR
- Over the medium run:
  - If anchored expectations: π goes back to initial value
  - If adaptive expectations: π constant at a higher value

The monetary policy has real effects in the short run only